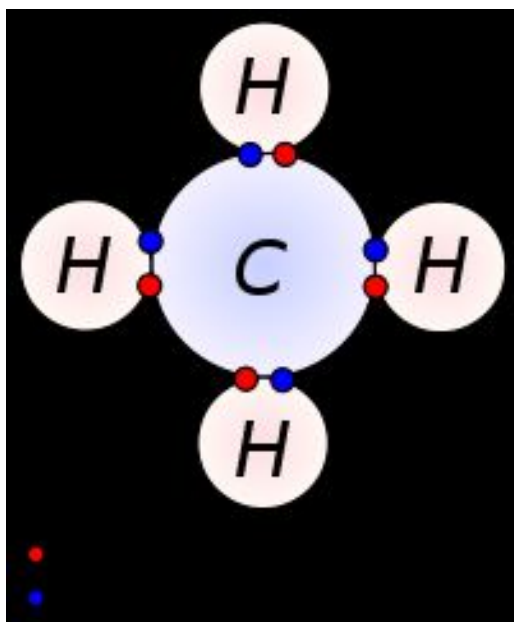


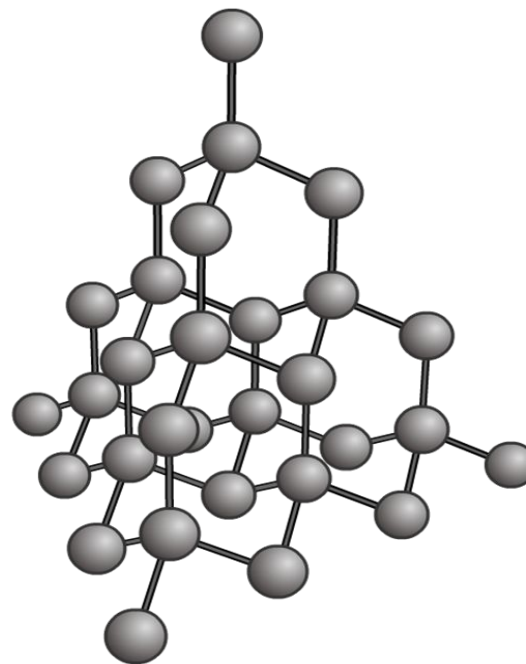
Giant covalent compounds.

To know the properties of giant covalent substances

What's the difference?



Simple



Giant

Graphite

Learning Objective:

- Describe the structure and bonding of graphite.
- Explain the properties of graphite.
- Explain the similarity to metals.

Keywords

delocalised electrons, graphite, hexagonal rings, weak bonds

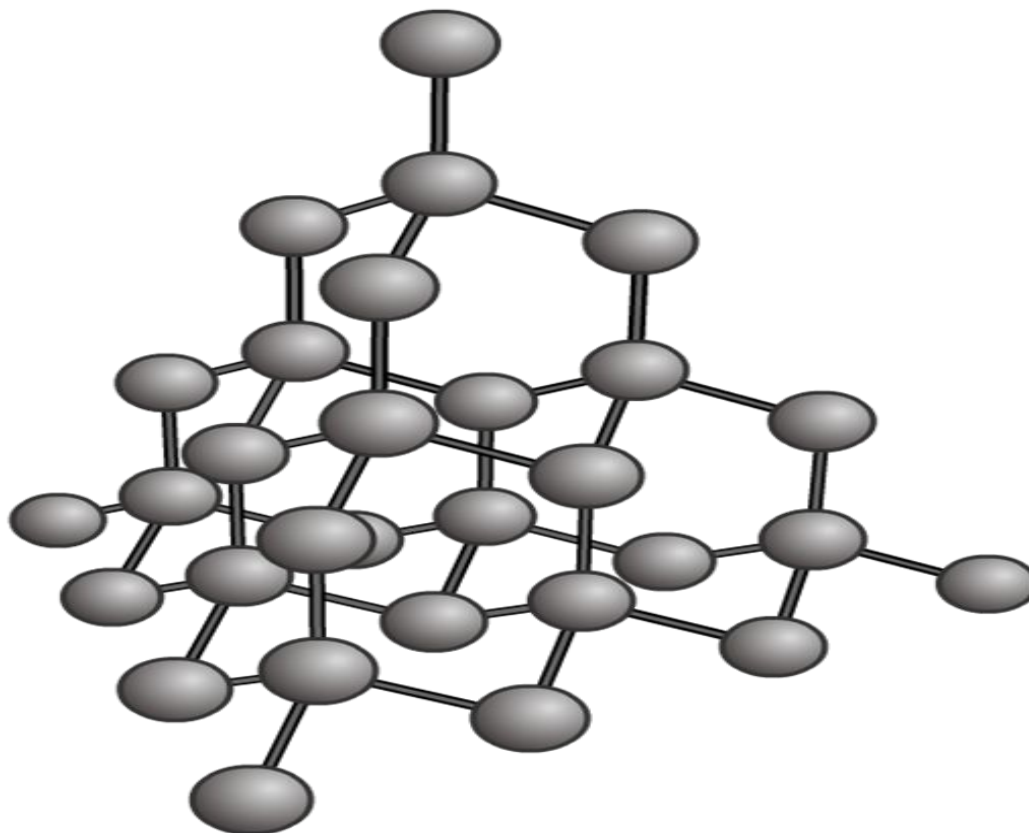


Queen
Elizabeth
School

Lesson objectives

Describe the properties of diamond and graphite, in terms of bonding and structure, and relate their uses to these properties.

Giant covalent compounds



What are giant covalent structures?

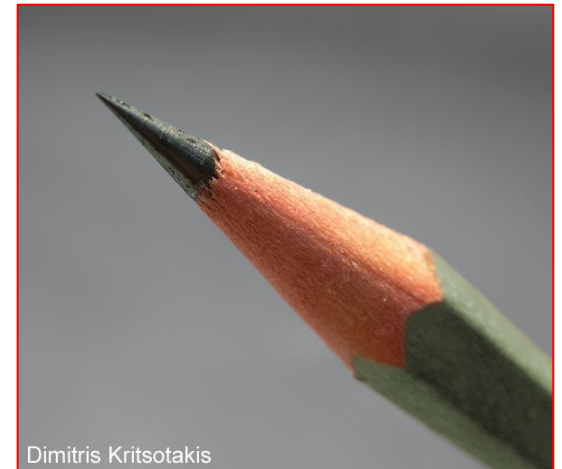
Sand



Diamond



Graphite



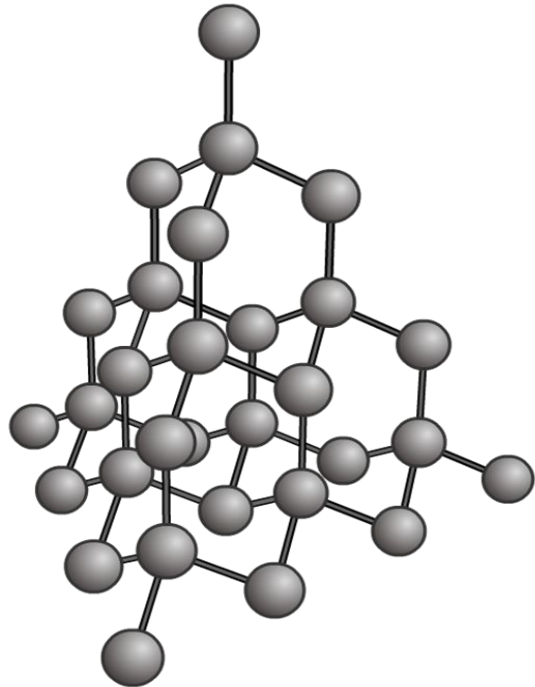
Giant covalent compounds.

To know the properties of giant covalent substances

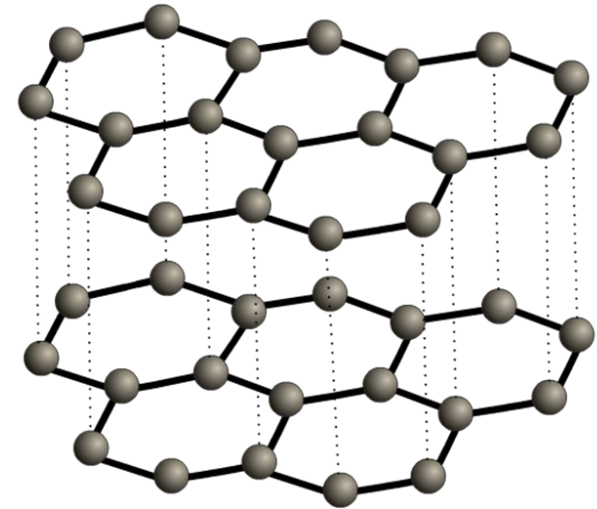
Both diamond and graphite are made up of carbon atoms.

Diamond and graphite appear to be very different substances but what do they have in common?



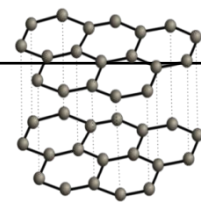
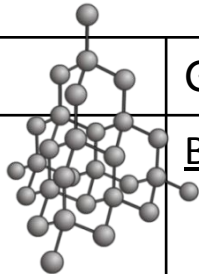


Diamond



Graphite

Diamond <u>Bonding & Structure</u> What is it made of? How are the atoms bonded? What forces hold the structure together?	Graphite <u>Bonding & Structure</u> What is it made of? How are the atoms bonded? What forces hold the structure together?
<u>Properties (appearance)</u>	<u>Properties (appearance)</u>
<u>Uses</u> What is it used for?	<u>Uses</u> What is it used for?

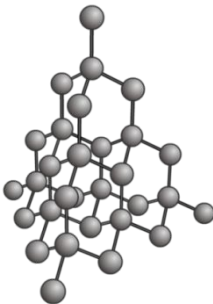
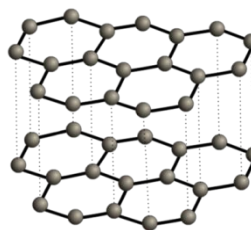


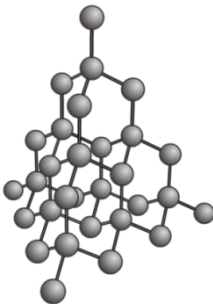
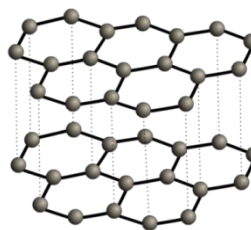
CLOSE READING

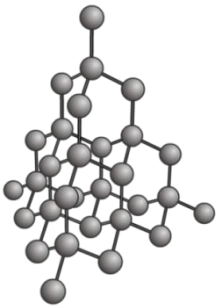
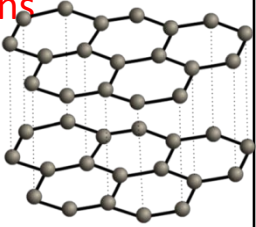


You pay close attention to the sentences, taking time to understand the meaning.



Diamond	Graphite
Bonding & Structure 	Bonding & Structure 
Properties	Properties
Uses	Uses

Diamond	Graphite
Bonding & Structure 	Bonding & Structure 
Properties	Properties
Uses	Uses

Diamond	Graphite
Bonding & Structure Uniform structure 1 carbon bonded to 4 others Strong forces between carbon atoms Giant covalent 	Bonding & Structure Bonded with 3 others carbons Form Layers Strong forces between the carbons Weak forces between layers Giant covalent 
Properties Hard Transparent Shiny High melting and boiling point Insulators	Properties Soft Slippery High melting point and boiling point
Uses Drill bits Jewellery	Uses Pencils Electrodes Lubricants

Quiz

1. *How many electrons does carbon have in its outer shell?*

4

Quiz

2. What do we call large structures that contain only covalent bonds?

Giant covalent molecules

Quiz

3. Name the two main Giant Covalent Structures.

Diamond and graphite

Quiz

4. In diamond, how many other carbon atoms is each carbon covalently bonded to?

1 carbon bonded to 4 other carbon atoms

Quiz

5. Why is graphite soft and slippery?

Layers and weak intermolecular forces
between the layers

Quiz

6. What property is the result of graphite's delocalised electrons?

The layers in graphite allow electrons to move freely between the layers

Quiz

7. What is a covalent bond? & What types of elements do they occur between?

Sharing of electrons between non metals

Quiz

8. What is the key property of diamond?

Hard

Giant covalent compounds.

To know the properties of giant covalent substances

Describe the properties of diamond and graphite, in terms of bonding and structure, and relate their uses to these properties. [6 QWC]

Sc- Plan (bullet points):

- ✓ Diamond- bonding, structure, uses and why used for that (look at properties)
- ✓ Graphite- bonding, structure, uses and why used for that (look at properties)

Giant covalent compounds.

To know the properties of giant covalent substances

Describe the properties of diamond and graphite, in terms of bonding and structure, and relate their uses to these properties. [6 QWC]

Diamond has ...

Each carbon atom joined....

The structure is...

This means its properties include....

Diamond can be used for...because...

Another use is...because...

Graphite has...

Each carbon atom is joined....

Again its structure is...but it is in....

The properties of graphite include ...

It can be used for...because...

It can also be used for...because...

Key content

Diamond

Bonding: uniform covalent / strong covalent bonds throughout / each carbon atom joined to four others

Structure: giant covalent

Properties and uses: hard – drill bits, transparent /sparkly – jewellery

Graphite

Bonding: strong bonds within layers and weak bonds between layers / carbon atoms / each joined to three others

Structure: giant molecular / covalent in layers

Properties and uses: soft – pencils, electrical conductor – electrodes

Key content

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Structure: giant molecular / covalent in layers

Properties and uses: soft – pencils, electrical conductor – electrodes

Simple and giant covalent compounds.

To know the properties of simple and giant covalent substances.

Describe the properties of diamond and graphite, in terms of bonding and structure, and relate their uses to these properties. [6 QWC]

Diamond has a uniform covalent bonds. Each carbon atom joined to four others. The structure is giant covalent. This means its properties include it being hard, transparent and shine, having a high melting and boiling point and not conducting electricity. Diamond can be used for drill bits because it is hard due to the strong bonds between the carbons.

Graphite has strong bonds within layers and weak bonds between layers. Each carbon atom is joined to three others. Again its structure is giant covalent, but it is in layers. The properties of graphite include that it is soft, slippery and has a high melting and boiling point. It can be used for pencils because it is soft due to the weak forces between the layers.

Simple and giant covalent compounds.

To know the properties of simple and giant covalent substances.

Success criteria-

- ✓ Identify general properties of simple covalent substances and giant covalent substances.
- ✓ Research the structure, properties and uses of allotropes of carbon.
- ✓ Describe and explain the properties of diamond and graphite, in terms of bonding and structure, and relate their uses to these properties. [6 QWC]

Simple and giant covalent compounds.

To know the properties of simple and giant covalent substances.

What 5 words summarize what we have learnt about covalent bonding today?

Reduce to 3.

Put all 3 into 1 sentence.

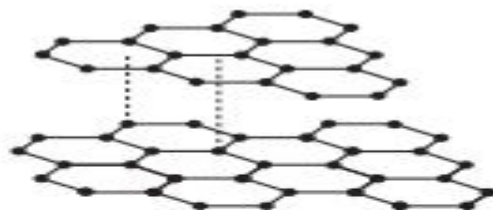
2. (a) Draw a line to link each substance with its structure.

[2]

One has already been done for you.

Substance		Structure
graphite		metallic
potassium		simple covalent
nitrogen		giant covalent
sodium chloride		giant ionic

(b) State which of the substances in part (a) has the structure shown by the following diagram. [1]



Substance

(c) Nitrogen is an example of an **element** with a simple covalent structure.

Name a **compound** with the same structure.

[1]

.....

- (c) Using the electronic structures below, draw a dot and cross diagram to show the bonding in a molecule of hydrogen peroxide. [2]

The structural formula for hydrogen peroxide is as follows.



hydrogen = 1

oxygen = 2,6

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The structural formula for hydrogen peroxide is as follows.



hydrogen = 1

oxygen = 2,6

Challenge

- WJEC additional
- Pg 104- Qus 13-18
- Pg 106- Qus 19-23